

1. Numpy: (Numerical Python)

It is a Python library, used for working with numbers and arrays.

It makes it easy to perform mathematical and scientific computations quickly and efficiently.

- i. *Numpy* provides the *ndarray (n-dimensional array)*, which can store large amount of numerical data.
- ii. It is much faster than Python List because it is optimized for numerical operations.
- iii. *Numpy* arrays are best for mathematical data or large datasets.

For using *Numpy* arrays, we need to download the *numpy* module in the Python environment using *pip (package installer for python)*.

```
pip install numpy
```

Numpy DataTypes:

<u>Data Types</u>	<u>What it Represents</u>	<u>Typical Use Case</u>
<i>int32</i>	<i>32-bit integer</i>	<i>Whole numbers (counts, IDs, indexing).</i>
<i>int64</i>	<i>64-bit integer</i>	<i>Larger integers (big datasets, IDs).</i>
<i>float32</i>	<i>32-bit decimal</i>	<i>Machine Learning (saves memory, faster training on GPUs).</i>
<i>float64</i>	<i>64-bit decimal</i>	<i>Default for Numpy, scientific calculations and high precision.</i>
<i>bool</i>	<i>True/False values</i>	<i>Masks, conditions, filtering.</i>
<i>complex64 / complex128</i>	<i>Complex numbers</i>	<i>Advanced math, signal processing.</i>
<i>string_</i>	<i>Text data</i>	<i>Labels, categorical data (but pandas is preferred).</i>